Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is that the server was overwhelmed with a large number of SYN requests. When this happens, the server sends **[RST, ACK]** packets, which results in the user receiving a timeout error.

The logs show that the server was normally completing a handshake with a legitimate employee, and from log number **52**, the attacker sends the first SYN, which the server responds to normally. When the attacker continued sending SYN packets, the server began to run out of capacity, until it completely ran out at log **125**.

This event could be: This event is caused by DoS attack from the ip address 203.0.113.0

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. The three steps of the handshake are:

1. Initially, the user sends a **SYN** request to the server using the TCP protocol.
2. The server then responds to that SYN by sending **[SYN, ACK]** back to the user.
3. When the user receives the **[SYN, ACK]**, an **ACK** is sent back as an acknowledgment.

This completes the three-way handshake. When the handshake is completed and the server is connected, the user receives the HTML page using the HTTP protocol.

When a malicious actor sends a large number of SYN packets all at once, the server becomes overwhelmed and runs out of capacity, causing it to stop responding or crash.

The logs indicate activity in three colors: green, yellow, and red, which help visualize the attacks and the normal user requests.